

MIDTERM Surface Tension, Curvature Elasticity, and Thermodynamics
(Original problem(s): do not repost anywhere without permission)

Problem 1: Struggles in Motion, or Mechanics underlying kinematics. (10) Please list all power laws of the form $d \propto t^n$ included in <https://www.youtube.com/watch?v=jv6IKH7aO5s>. List in each case the pair of mechanical quantities that provide the basis for the observer's power law (or kinematics). Feel free to refer to the article itself: <https://pubs.rsc.org/en/content/articlehtml/2024/sm/d4sm00263f>

Problem 2: Phase behavior and models derived using assumptions included in Flory-Huggins Theory (30)

Read carefully the book chapter on phase behavior and thermodynamics from the textbook on *Polymer Physics* by Rubinstein and Colby. Answer the following questions: (Graduate: 5+5+10+10=30; UG 10+5+15, with d optional)

- Rederive 4.25 (Flory-Huggins expression), listing all the assumptions and a critical examination of the consequences of the assumptions (and approximations).
- What is the solubility parameter? How does it compare with the χ parameter?
- In Section 4.5, the free energy expression for low composition is discussed, and a definition for osmotic pressure is provided. Rederive the virial expansion form (equation 4.71) and comment on the relationship between the second virial coefficient and the Flory-Huggins chi-parameter.
- Using expressions 1 and 2 as defined in Spruijt *et al.* (*Macromolecules* 2010), derive the virial expansion form, and comment on the relationship between the second virial coefficient and charge effects.

Problem 3: Curvature elasticity and Morphology of Vesicles (25)

Read the book chapter by Seifert and Lipowsky carefully, and also peruse through any additional notes on the topic in the folders on the blackboard. You can revisit the video we referred to in class for insights or help: https://youtu.be/VAe2f_Q-Prg?si=IAucjoR-j6Y7Bc9i Answer the following questions: (Graduate: 5+10+10; UG 10+15, with c optional)

- With reference to the spontaneous curvature model (see equation 7): Define each term that contributes to the free energy expression. Provide the S.I. units and the typical values for the material properties (numbers for up to three lipids), and also describe how these are measured, problems & controversies therein. (Feel free to cite and quote literature included in the notes and any additional papers).
- Is it correct to say that the model provides an incomplete description of the bilayer free energy as no mention is made of the bilayer architecture. How would you assess the role of bilayer architecture, and does it require you to seek and define more material parameters? If yes, list those.
- Based on your understanding of different terms that contribute to surface tension and curvature elasticity, assess what material properties determine shape, size, curvature and even coalescence dynamics in emulsions. Include an example or two from literature of such studies (include citation to the references).

Problem 4: Pendant Drop Analysis (20)

Critique the paper on "Isotherms of phospholipid monolayers measured by a pendant drop technique" by Li *et al.* (1996). What are the equations used for pendant drop analysis? Do you expect there to be additional effects (curvature elasticity, interfacial stresses, etc.)? No more than a couple of pages, including references or schematics.

Problem 5: (Already submitted: your A4 sheet with formulae) (15)

Q1),

Constant Velocity: $d \propto t$ $n=1$

In the video it stated: a cart moving at a steady speed.
dominant is velocity

- the mechanical pair is d, v

v is constant.

written as L/T

Constant acceleration, $d \propto t^2$
 $n=2$

In the video the example is

free fall as the fall last

longer the fall accelerate.

dominant is acceleration.

mechanical pair d, a $a = \text{const}$

$$d = \frac{1}{2} t^2 \quad (= d = a t^2)$$

Constant Jerk: (Smooth motion.

" $d \propto t^3$ " $n=3$

In the video this is found in controlled movement in mechanics like robots.

mechanical pair $d, j \Rightarrow j = \frac{da}{dt}$ $n=1/2$

Diffusion (randomized motion) $d \propto t^{1/2}$

the example in the video is die diffusing in water over time. "Brownian motion".

mechanical pair: $D = L^2/T$, $d = Dt^{1/2}$

Subdiffusion (viscoelastic system)

$$d \propto t^n$$

$$n < 1/2$$

In the video the example for this is motion of polymer, non-newtonian fluids.

The mechanism of this is d with mechanical interference from friction.

Super diffusion: (Turbulent flow)

$$d \propto t^n, \quad 1/2 < n < 2$$

In the video the example is turbulent flow.

The mechanism for this is d with motion $\propto t^n$, or randomness,

Question 2:

a) Rederive eq 4.25, List assumptions and consequences of assumptions and approximations:

Eq 4.25:

$$\Delta F_{mix} = kT \left[\frac{\phi}{v} \ln \phi + (1-\phi) \ln(1-\phi) + \chi \phi(1-\phi) \right]$$

Solution to A:

Start w/ Lattice Polymer + Solvent mixture.

$$\overset{\text{polymer}}{\phi} = \frac{n_p M}{n} \quad \text{and} \quad \overset{\text{solvent}}{1-\phi} = \frac{n_s}{n}$$

Theory: 1 solvent or 1 polymer

Now we must apply Entropy mixing to expand the equation $n_{s,p}$ as a place:

Using chains as a single count for a assumption.

Configurations

$$\Omega = \frac{M!}{n_s! (n_p N)!} = \text{Chain configurations}$$

With the configuration applied we can use Flory's approximator.

$$\Delta S_{mix} = -k \left[n_s \ln \frac{n_s}{n} + n_p \ln \frac{n_p}{n} \right]$$

rewrite by converting volume fractions:

$$\frac{\Delta S_{mix}}{k} = -n \left[\frac{\phi}{v} \ln \phi + (1-\phi) \ln(1-\phi) \right]$$

With The mixing Set up we can
appt + ΔH_{mix} and enthalpy mix,

Possible mixing options

$E_{PP} = \text{Polymer} + \text{Polymer}$

$E_{SS} = \text{Solvent} + \text{Solvent}$

$E_{SP} = \text{Solvent} + \text{Polymer}$

Next we assume contacts as
a mean approximation

$$C_{\text{contacts}} \approx 2 M \phi (1 - \phi)$$

with this we Define the
Flory Parameter: χ

$$\chi = \frac{z}{kT} \left(E_{PS} - \frac{1}{2} (E_{PP} + E_{SS}) \right)$$

Then we combine to make the
enthalpy mix

$$\Delta H_{\text{mix}} = kT M \chi \phi (1 - \phi)$$

with Enthalpy mix established
we can add the total free energy
to calculate Derivatives

total energy eqn.

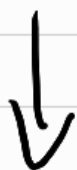
$$\Delta F_{\text{mix}} = \Delta H_{\text{mix}} - T \Delta S_{\text{mix}}$$

appt + to Flory parameter and

factor kT out to get

let's cite to make Flory eqn.

$$\frac{\Delta F_{\text{mix}}}{kT} = \frac{\phi}{N} \ln \phi + (1 - \phi) \ln (1 - \phi) + \chi \phi (1 - \phi)$$



ASSUMPTIONS Made. For Developing
The equation:

Assumption 1: The Lattice model
has identical sites, and volume.
Not realistic in polymer science.

Assumption 2: The mixture is
incompressible, this assumes that
there are no voids i.e. " $\phi + (1-\phi) = 1$ "

Assumption 3: Polymer as connected segments
assuming chain occupies N sites

Assumption 4: The assumption that
random mixing as mean is not
realistic. No accounting of local correlations.

Assumption 5: Polymer segments are
all equal in size (polymers + solvents)
sizes are all random.

Assumption 6: The equation does not
account for stiffness, assuming
that it is all flexible and
fixed in geometry.

Assumption 7: Equation only accounts
for nearest neighbor interactions, Not
all.

Assumption 8: χ is only used
to measure only as a single capture
for all energies.

Assumption 9: χ does not factor in
the composition of mixture.

Assumption 10: Assumes that all
interactions are pairwise.

Consequences of all assumptions...

Entropy is assumed as $\frac{1}{N}$, this assumes that all entropy is from mixing. Causing, eqs + phase separation, and defects on χ . This is why local correlations and chain permutations.

Mean field assumption simplifies mixing, the temperature critical pts and phase boundaries are not features, this cannot capture concentration changes and separation in the micro phase.

χ -parameter is simplified as a constant. When it must be a function of $\chi = \chi(T, \phi)$, this has no proper phase diagrams, and incorrectly predicting hard behavior.

Lattice and size are all equal. as polymer is the same as a solvent. This does not factor in size mismatch and non-elliptical and compressible systems.

No compressibility is also assumed. Saying that volume is fixed and not describe pressure effects, chain order, and liquid behavior.

No chain structure effects are also assumed, this ignores stiffness, branching and structure. This leads to no difference in linear and branched polymer chains and forgets about entropy.

another assumption is that contacts was simplified to nearest neighbor, forgetting about other bonds and interactions.

b.) What is the Solubility Parameter compared to χ

The Solubility parameter is

$$\delta = \sqrt{\frac{\Delta E_v}{V_m}} \quad \begin{array}{l} \Delta E_v, \text{ Cohesive energy} \\ V_m, \text{ molar volume} \end{array}$$

The Solubility parameter can be defined as forces within attraction forces, this allows to define if it separates easily or sticks together.

Usually when Solubility is High the force of Bonds are polar or H and do not separate easily, while weak Solubility (δ) are non polar and break off easily.

Know w/ Flory Huggins this can be applied to the Bonds conditions for mixture favorability, with $\chi > 0$ being Unfavorable and $\chi < 0$ Being favorable.

These two equations can be combined as the following as given in the book for non polar polymers.

$$\chi = \frac{V_s}{RT} (\delta_p - \delta_s)^2$$

where
 V_s = Solvent vol
 δ_p, δ_s = Solubility of solvent and polymer.

Comparing χ and δ can be done by comparing the nature of how both work with material Bonds and interaction. Since δ is cohesion of a substance, and χ is energy needed for separation, both can depend on temperature as a better indicator. Although χ depends on more comprehensive molecular strength, δ depends on empirical and thermodynamic properties while χ is a thermodynamic analysis.

The Other Aspect of Comparing χ and δ is
 how both are used for polymerization.
 δ is non polar and for specific
 interactions, while χ is solubility parameter.

The relationship $\chi \propto (\delta_p - \delta_s)^2$

(.) re derive from (4.71).

$$(4.71) = \Pi = \frac{kT}{b^3} \left[\frac{\phi}{N} + (1-2\chi) \frac{\phi^2}{2} + \frac{\phi^3}{3} + \dots \right]$$

To start the Derivation of Free
 energy:

$$\frac{\Delta F_{mix}}{kT} = \frac{\phi}{N} \ln \phi + (1-\phi) \ln (1-\phi) + \chi \phi (1-\phi)$$

Treating osmotic pressure Π as
 limit of dilution $\phi \ll 1$

now we derive osmotic pressure

$$\Pi = \frac{1}{V_s} (N_s - N_s^0)$$

appt lattice:

$$\Pi = \frac{kT}{b^3} \ln a_2$$

appt Flory Huggins:

$$\ln a_2 = \ln(1-\phi) + \phi + \chi \phi^2$$

now we expand the series

$$\ln(1-\phi) = -\phi - \frac{\phi^2}{2} - \frac{\phi^3}{3} - \frac{\phi^4}{4} - \dots$$

appt lattice (av):

$$\ln a_2 = \left(-\phi - \frac{\phi^2}{2} - \frac{\phi^3}{3} - \frac{\phi^4}{4} \right) \phi + \chi \phi^2$$

Simplify by cancel terms

$$-\phi + \phi = 0$$

This makes: $\ln a_s = (\chi - \frac{1}{2})\phi^2 - \frac{\phi^3}{3} + \frac{\phi^n}{n}$

now we combine w osmotic pressure.

$$\Pi = \frac{\pi T}{b^3} \ln a_s$$

into: $\frac{\Pi}{\pi T} = \frac{1}{b^3} \left[(\frac{1}{2} - \chi)\phi^2 + \frac{\phi^3}{3} + \frac{\phi^n}{n} \right]$

now we apply Ideal polymer enter,

$$\Pi = \frac{1}{b^3} \left[\frac{\phi}{N} + (\frac{1}{2} - \chi)\phi^2 + \frac{\phi^3}{3} + \frac{\phi^n}{n} \right]$$

now we separate Π to set eq 4.71

$$\Pi = \frac{\pi T}{b^3} \left[\frac{\phi}{N} + (\frac{1}{2} - \chi)\phi^2 + \frac{\phi^3}{3} + \frac{\phi^n}{n} \right]$$

The virial coefficient can be given

The equation can be given

$$\Pi = \pi T (C + B_2 C^2 + B_3 C^3 + B_n C^n)$$

Setting C as the coefficient $C \sim \frac{\phi}{b^3}$

1st term: The 1st term is Ideal $\frac{\phi}{N}$

2nd virial term: B_2 & $(\frac{1}{2} - \chi)$

with that equation we can demonstrate.

That when $\chi < 1/2$, $B_2 > 0$

is a good solvent, with strong repulsion forces, from polymer expansion

We also see that when

$\chi = 1/2$, $B_2 \geq 0$, This is Ideal
Cmn behavior, with no interaction

and when $\chi > 1/2$, $B_2 < 0$
we find attractive interaction, thus
phase separation.

d). Derive the virial expansion
form;

using required equation for Polym.

$$\Pi = \Pi_{\text{ideal}} + \Pi_{\text{int}} + \Pi_{\text{elec}}$$

Now we make it virial expansion in
Lattice form

$$\Pi = kT (C + B_2 C^2 + B_3 C^3 + B_n C^n)$$

convert to Lattice

$$\Pi = \frac{kT}{b^3} \left[\frac{\phi}{N} + A_2 \phi^2 + A_3 \phi^3 + A_n \phi^n \right]$$

now we expand Π_{ideal}

$$\Pi_{\text{ideal}} = \frac{kT}{b^3} \frac{\phi}{N}$$

expanding Π_{int}

$$\Pi_{\text{int}} = \frac{kT}{b^3} \left(\frac{1}{2} - \chi \phi^2 + O(\phi^3) \right)$$

expanding Π_{elec}

$$\Pi_{\text{elec}} = \frac{kT}{b^3} \left(f^2 \phi^2 + O(\phi^{5/2}) \right)$$

Combine the expression.

$$\Delta = \frac{kT}{b^3} \left[\frac{\phi}{N} + \left(\frac{1}{2} - \chi + \alpha f^2 \right) \phi^2 + \phi^2 + O(\phi^3) \right]$$

The coefficient is:

$$B_2 = \left(\frac{1}{2} - \chi \right) + \alpha f^2$$

What this does is it adds a repulsive application along the chain. through electric charge, to achieve

The charge we need to use salt to apply the charge. This means the following parameters.

$$B_2 = 0, \quad \frac{1}{2} \chi + \alpha f^2 = 0$$

With the charge applied we can have repulsion, attraction through charges from salt

Q3

a.) eq 7

$$F_{sc} = \left(\frac{\kappa}{2}\right) \oint dA (2H - C_0)^2 + \kappa_s \oint dA \kappa$$

Given Constants:

κ is Bending rigidity

This used to measure resistance to mean-curvature bending.



values:

DOPE	20
DMPC	10-15
DPPE	40-60

units

Units are in J

Typically found in Phospholipid-bilayers
 $\sim 10^{-19}$ J

H is Mean Curvature.

Equation: $H = \frac{1}{2} \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$

Units are: m^{-1}

C_0 is Spontaneous Curvature.

This represents the preferred mean curvature due to membrane asymmetry. This can come from environmental effects on the bias of membrane structure composition.

Units are: m^{-1} $C_0 = 10^{-6}, 10^3, 10^7$

k_g is Gaussian bending rigidity.

This is a multiplier for Gaussian Curvature. the k_g is usually seen as order of magnitude as k_c at approx: 10^{-19}

Units are in J.

k_c is Gaussian Curvature

equation: $k_c = \frac{1}{R_1 R_2}$ -10, -15, -20

Units are m^{-2}

dA is Surface - area

Units are m^2 .

To measure these parameters.

To measure Bending rigidity (k_c) can be done in 3 methods.

The first method is mechanically to do this we can measure pull force and do a force balance for k_c .

This works as it allows for mechanical measuring and interpretation of force.

Another method to solving this is calculating the thermal

fluctuation, this mostly uses a few non-standard shapes like quasi-spherical vesicle.

Tex work well in secret membranes. But tend to require a lot of assumption on the model.

The next method to look at is entropic tension.

This combines the last 2 methods measuring membrane force and thermal energy. The issue is it takes on the model.

To measure Caution rigidity (κ_s) the method to measure is a force on a topological variant that is fixed in shape to not interfere with the shape minimizer. The issue in using is the difficulty in measuring. Since it's parameters are difficult to measure in vesical shapes.

Ability to measure Spontaneous Curvature (C_0), a different method can be used on its asymmetric leaflet composition, and in its external conditions... the complication comes from fitting slope as it is too much to measure.

b). The Springer - Curvature model provides an incomplete description in bilayer free energy analysis, since when examining Eq. 7,

The membrane is assumed to be a perfect structure but lacks insight in the monolayer parameter.

The lack of insight forgets about bending impacts in the inner and outer parameters. This bilayer is critical in examining the cell elastic quality, causing issues with inconsistency in elastic measurements throughout the model.

The parameters included apply equilibrium in a leaflet are "DA". To demonstrate the non-local bending rigidity (κ), this also applies to monolayer separation and leaflet separation.

Javier Silva

4/22/2026

CHE 457

Dr. Sharma

Midterm question 4 solution:

Analytical critique of the paper “Isotherms of phospholipid monolayers measured by a pendant drop technique” Li et al. (1996) .

In the paper “Isotherms of phospholipid monolayers measured by a pendant drop technique” by Li et al the focus of research is on examination of if Isotherms by pendant drops. The method determines the surface tension of the liquid by the shape of the interference. For this experiment specifically the method is called Axisymmetric drop shape analysis (ADSA), this measured the surface pressure are of the following phospholipid monolayers DPPC and DMPE. The data from the experiment is then compared to the conventional method Langmuir – Blodgett (LB) trough, The LB trough method also measures the Surface pressure area isotherms. Both methods are generally good in being used in the experiment, but notable discrepancies are shown by the mostly in the coexistence phase region. These discrepancies are the result of the physical limitations specifically in the pendent drop method by Li.

To explain the limitations of Li's method we must establish the governing equation that creates the foundation of the theorem behind the experiment.

First, we establish the isotherm equation that demonstrates the surface tension leading to the drop of formation from surface pressure:

$$\pi(A) = \gamma_0 - \gamma \text{ (Eq. 1)}$$

$\pi(A)$ = isotherm

γ = surface tension

γ_0 = initial surface pressure (turns into $\gamma_{control}$)

This defines overall theorem of the experiment.

With the isotherm equation established the Young-Laplace equation can be used to set up the core of the derivation:

$$\Delta P = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \text{ (Eq. 2)}$$

ΔP = Pressure difference across the interface of the drop

γ = surface tension

R_1, R_2 = The radii of curvature of the drop

This defines the shape of the drop.

Since the Young-Laplace equation can be combined with Hydrostatic Variation to make a differential equation for the shape:

$$\Delta P(z) = \Delta P_0 + \Delta \rho g z \text{ (Eq. 3)}$$

$\Delta \rho$ = Difference in density from liquid and air interactions

g = Gravity

z = Vertical coordinate

This allows for the measurement of gravity and capillary length.

With the establishment of a coordinate, we can now apply the Axisymmetric drop shape equation into the differential equation of the shape:

$$\frac{d\psi}{ds} = \frac{\Delta \rho g z}{\gamma} - \frac{\sin \psi}{r} \text{ (Eq. 4)}$$

With the following coordinates for the drop:

$$\frac{dr}{ds} = \cos \psi \text{ (Eq. 5)}$$

$$\frac{dz}{ds} = \sin \psi \text{ (Eq. 6)}$$

s = Arc length

ψ = Tangent angle

r, s = Coordinates of the drop

This allows the ADSA to be model with γ via nonlinear regression.

With the basis of equation fully established for the ADSA experiment we can now rewrite the entire equation for the surface pressure and isotherm:

$$\pi(A) = \gamma_{control} - \gamma(A) \text{ (Eq. 6)}$$

This allows the equation to measure thermodynamically instead of geometrically.

The pendent drop also can be turned into a dimensionless number to ease the transfer of units through the bond number:

$$Bo = \frac{\Delta\rho g z}{\gamma} \text{ (Eq. 7)}$$

This equation is now used to measure the results of the experiment and compared to the LB analysis.

Now on to the evaluation of if ADSA is a viable alternative to LB analysis, The method developed by Li is capable of producing the phase transitions found in the LB Troughs which allows for the method to be a good alternative in consideration for measuring surface tension. Although there are some deviations found in using ADSA due to the limitations of the derived equation.

To further expand on the analysis of the equation, the strength of the ADSA method is the ability of the equation used in the experiment being able to couple geometric parameters and measurements with thermodynamic parameters and measurements. The ability of the equation to do that makes it easy to cross analyze data in geometric and thermodynamic units. This is further simplified with the introduction of the dimensionless Bond number that combines both properties improving the analysis. This ADSA outperforms the LB method when it comes to thermodynamic properties. Another strength of the ADSA method when compared to the LB method is the small sample volume of the drop that

allows for more consistent temperature allowing for more uniform results in thermodynamics. Finally, ADSA outperforms the LB method through the direct measurement of γ instead of the process needed to measure in the LB method. This allows for more ease in ADSA to measure the surface pressures when they are close to the collapse of the drop. This is simpler than LB making it more efficient in analyzing in those situations the conventional method that the LB method does.

The ADSA method does have its limitations that does not allow it to be a viable method to the conventional LB method. The fundamental flaw in this method is that it does not fully account for effects of curvature, the writer assumes that the pendent drop is equivalent to a flat surface which is fundamentally flawed as it does not account for effects of curvature such as the tension from the line, the properties of the shape and most importantly the behavior of the coexisting phases. The paper correctly states that there are limitations of the method but still do not justify its ability to replace the LB method. (Li 1996)

The next issue is the non-equilibrium compression that is the result of kinetics and not of equilibrium. The ADSA method has faster compression rates at (1.3–6.1 $\text{\AA}^2/\text{s}$) (Li et al 1996) than the LB method compression rates of (0.06–0.53 $\text{\AA}^2/\text{s}$). (Langmuir 2019) This leads to the issues of shifts in the coexistence regions that relate to the first issue of the experiment discussed earlier in critique. This common in alternative methods conducted by other researchers as the shift in coexistence phase can be altered easily by pressure. (Langmuir 2019)

A further limitation of the pendant drop method cannot fully decouple thermodynamic behavior from morphology and kinetics. Li et al. shows that in the coexistence region, monolayer structure is highly sensitive to compression conditions, with significant changes in domain size, shape, and distribution at different rates. Especially on a curved interface, these domains do not reflect equilibrium phase separation but are influenced by nucleation and diffusion, leading to non-equilibrium conditions. This makes interpretation difficult because the measured surface pressure area isotherm becomes a combination of intrinsic thermodynamics and structure. This leads to ADSA measurements in the not being able to represent a true equilibrium equation of state, but instead reflect transient state, focused on thermodynamic parameters.

A final key limitation of the pendant drop (ADSA) technique is the sensitivity to impurities and molecular area; both can significantly impact the accuracy of measured isotherms. This is due to ADSA using a very small droplet, which can cause impurities to remain concentrated at the interface. This affects surface tension and surface pressure, especially non-equilibrium conditions. Li et al. note that this effect is more pronounced than in Langmuir trough experiments (Langmuir 2016), where the larger subphase allows impurities to disperse. Contamination can also influence phase behavior, disturb coexistence regions, and interrupt isotherm slopes. Meanwhile, the molecular area is not directly measured but inferred from drop geometry and the amount of deposited material. This introduces additional uncertainty, as small errors in volume, concentration, or shape fitting propagate into the calculated area per molecule. As stated by Li et al., this can lead to systematic shifts along the axis when comparing ADSA with conventional

measurements . Since phospholipid phase transitions occur in narrow ranges of the molecular area, allowing for minor inaccuracies to interfere the precise location and nature of transitions, limiting the quantitative reliability of the technique .

The pendant drop technique, done through (ADSA), provides a method for determining surface tension and constructing surface pressure–area isotherms of phospholipid monolayers. Its ability to extract thermodynamic values from drop geometry, and its advantages such as small sample requirements and uniform experimental conditions, make it a possible alternative to LB measurements. As shown by Li et al., ADSA successfully captures the general features of monolayer phase behavior and agrees well with conventional methods in many regions . However, limitations are shown especially in the phase coexistence region from the technique rather than experimental error. Factors such as surface curvature, faster compression rates, and the coupling between domain morphology and non-equilibrium kinetics influence the measured isotherms, sensitivity to impurities and issues in molecular area affect the accuracy of the method. As a result, ADSA measurements combine thermodynamic and kinetic effects rather than developing an equilibrium equation of state. Therefore, I believe method is highly useful for specific applications like the phase state and measuring the max amount of pressure before collapse but needs work in order to fully be able to remove or minimize its limitations of the experiment.

References

1. Li, J. B.; Miller, R.; Vollhardt, D.; Weidemann, G.; Möhwald, H.

Isotherms of phospholipid monolayers measured by a pendant drop technique.

Colloid Polym. Sci. 1996, 274, 995–999. (Li 1996)

2. Zuo, Y. Y.; Chen, R.; Wang, X.; Yang, J.; Policova, Z.; Neumann, A. W.

Phase transitions in dipalmitoylphosphatidylcholine monolayers.

Langmuir 2016, 32, 8501–8506. (Langmuir 2016)

3. de Oliveira, F. O.; Tamashiro, M. N.

Phase transitions in phospholipid monolayers: Theory versus experiments.

Langmuir 2019, 35, 3848–3858. (Langmuir 2019)